
TOPIC 3

Changing ecosystems

Topics

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This topic looks at:

- the characteristics of two contrasting ecosystems, Antarctica and tropical rainforests
 - the threats that each faces and the strategies used to manage them.
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3.1 The characteristics of the Antarctic ecosystem

This chapter will explain:

- ★ the location of the Antarctic ecosystem
- ★ the climate of Antarctica
- ★ the features of the Antarctic ecosystem.

The location of the Antarctic ecosystem

The Antarctic refers to everything south of the Antarctic Circle ([Figure 3.1](#)) including land, ice shelves and sea, while Antarctica refers to the landmass that is the southern polar continent. Antarctica consists of two parts, East and West Antarctica. The landmass is about 14 million km², which is about 6 million km² larger than the USA. The Antarctic ice sheet covers about 98 per cent of the continent and is on average about 1.6 km thick. Antarctica contains around 90 per cent of the world's ice and 70 per cent of its fresh water. If all the ice in Antarctica were to melt, global sea levels could rise by up to 60 m.



▲ **Figure 3.1** Antarctica and the Antarctic Circle

A large part of the West Antarctica Ice Sheet (WAIS) lies below sea level and its edges are floating ice shelves. The WAIS accounts for about 10 per cent of the total volume of the Antarctic ice sheet, about 2 million km³. If WAIS were to melt, sea levels could rise 3–4 m. The East Antarctica Ice Sheet (EAIS) rests on the Antarctica landmass; it's a thicker, larger ice sheet, and is nearly 5 km deep.

In 2024 it was reported that scientists had identified a tipping point for the loss of ice sheets in Antarctica that could cause a greater rise in sea level than previously expected. They examined how warming gets between coastal ice sheets and the ground they rest on. The warm water melts some of the ice, allowing more water to flow in. This reinforces the warming. Scientists had warned in 2022 that the Greenland ice cap was in danger of collapsing, and in 2023 they found that accelerated melting in western Antarctica was inevitable for the rest of the century.

Interesting note

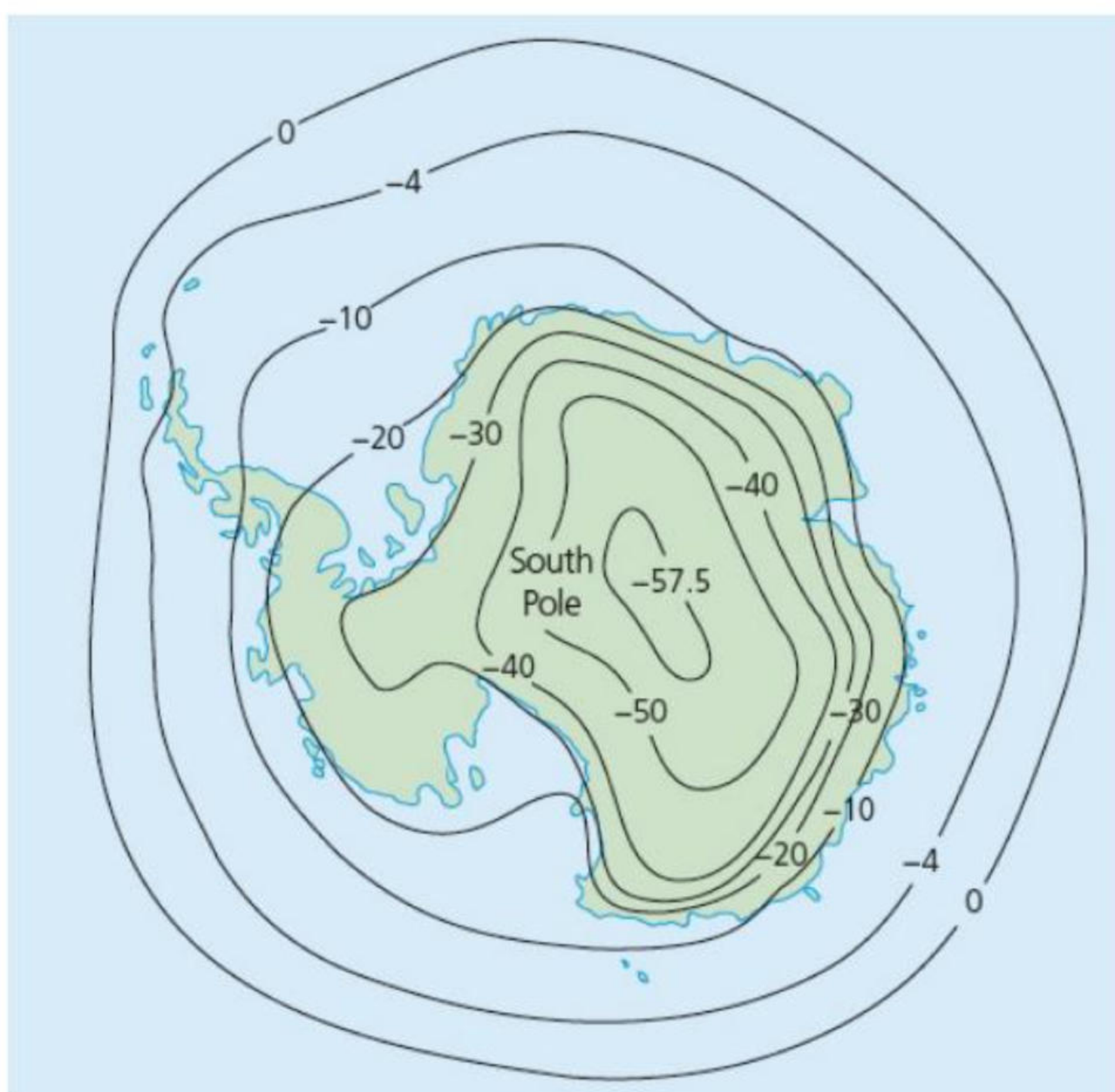
Antarctica is unique among the continents in that it is the only one with no indigenous population. The first 'indigenous' child to be born there was an Argentinian boy, Emilio Marcos Palma, born in 1978. Since then, more than ten children have been born in Antarctica. It is inhabited by around 5000 people in summer (scientists and support staff) and about 1000 people in winter.

The climate of Antarctica

Temperature

The main characteristic of Antarctica's climate, in terms of both duration and intensity, is that it is cold. On ice sheets in the interior, the winters are severe and the summers non-existent. However, in coastal areas, the winters are warmer and the summers cooler. There are many reasons why Antarctica is so cold:

- » Due to high **air pressure**, the continent is relatively cloud-free and receives little **rainfall**. High pressure areas have descending air so there is limited uplift or **cloud** formation. This allows much of the outgoing long-wave radiation to escape from the Earth's surface, making the region very cold.
- » Due to its high latitude, the low angle of the sun in relation to the ground means it receives less radiation. In the brief summer, however, there is sufficient solar radiation to melt snow and ice and to support life.
- » Due to its colour, Antarctica reflects much solar radiation. This reflectivity is called **albedo**. Dark surfaces absorb more solar radiation (up to 90 per cent), whereas fresh snow may absorb as little as 10 per cent and reflect the other 90 per cent.
- » There is a lack of dust and water vapour in the atmosphere. Dust and water vapour are important for trapping long-wave radiation from the Earth's surface, which heats the Earth's atmosphere. Hence, limited amounts of long-wave radiation are trapped over Antarctica to heat up the atmosphere.



▲ **Figure 3.2** Mean annual surface temperatures in Antarctica (°C)

Antarctica's temperatures are low throughout the year ([Figure 3.2](#)). The mean annual temperature is below freezing everywhere south of 60°S. On the continent, the mean annual temperature varies from about -60°C inland to around -10°C by the coast. There are seasonal contrasts too. On the continental interiors, temperature ranges from about -70°C in July to -25°C in January. In contrast, along the coast it ranges from -40°C in winter to -2°C in summer. The height of much of Antarctica also leads to lower temperatures. In general, temperatures fall 1°C per 100 m. Around 50 per cent of Antarctica is over 2000 m and 25 per cent over 3000 m. If those areas were at sea level, Antarctica would be 20–30°C warmer!

Winds are particularly important in Antarctica and can intensify the chilling effects of low temperatures ([Table 3.1](#)).

▼ **Table 3.1** The effect of wind speed on temperature – the wind chill effect

Air temperature (°C)	Wind speed (metres per second)				
	Calm	2.5	5	10	20
0	19	0	-7	-12	-18
-5	16	-6	-13	-19	-26
-10	13.5	-11	-19	-26	-33

-20	9	-21	-31	-40	-44
-30	4.5	-31	-43	-54	-63
-40	0	-41	-55	-68	-78

Blizzards are also a feature of Antarctica. When surface winds reach 10 m/s^{-1} , snow is picked up and saltated (bounced) along the ground. When winds reach a speed of 15 m/s^{-1} , they can pick up enough snow to reduce surface visibility to zero.

Precipitation

Precipitation levels in Antarctica are low and consist mainly of snow. Many of the central parts of East Antarctica receive less than 50 mm of water equivalent (that is, how much water the snow would produce if it fell as rain), whereas coastal areas can receive 200–600 mm. Once more, **altitude** has an impact. As moist air is forced to rise, there is condensation, cloud formation and precipitation, and maximum snowfall occurs at about 1600 m.

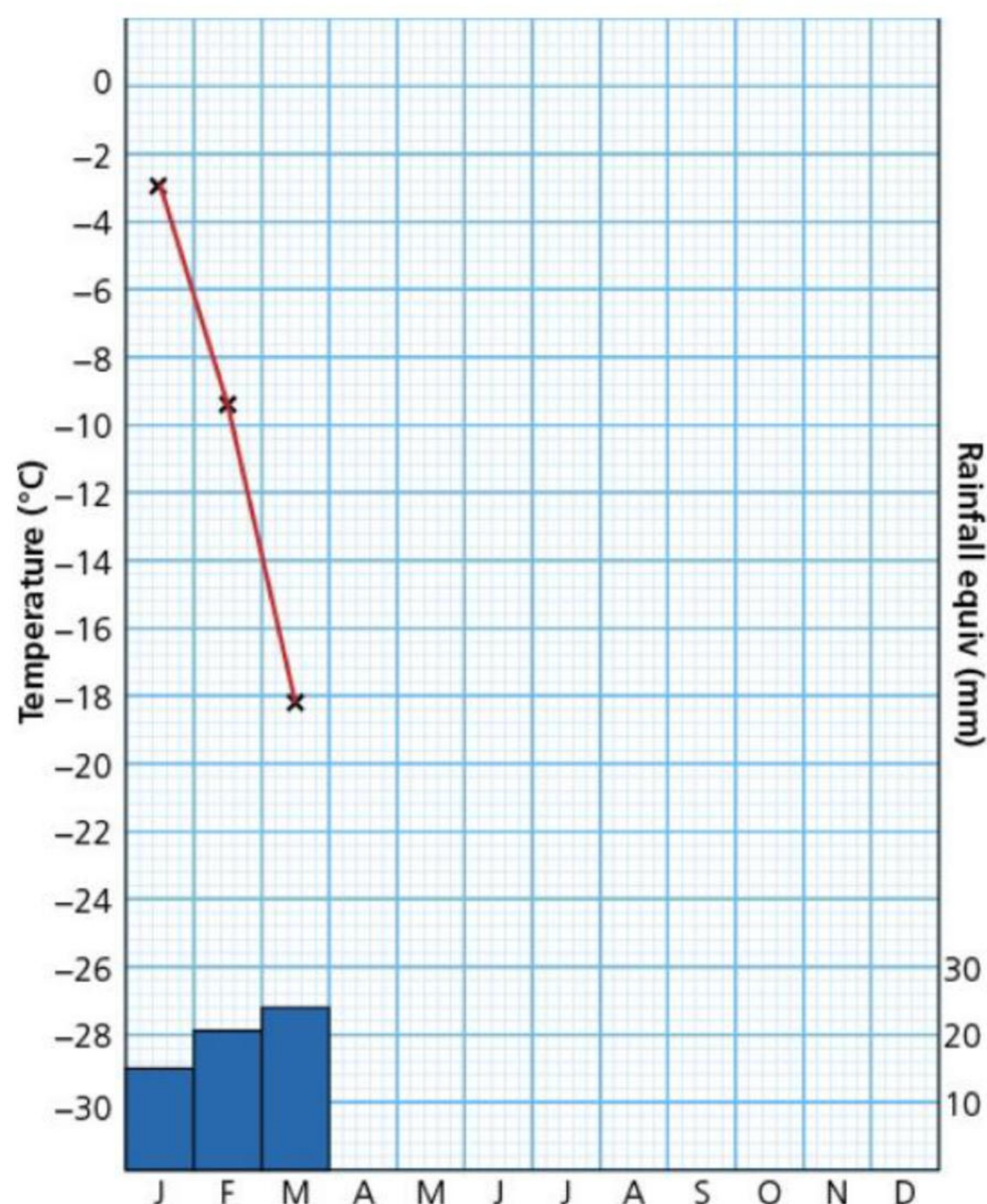
Activities

- Using [Table 3.1](#), state the wind chill when:
 - air temperature is 0°C and wind speed is 5 m/s^{-1}
 - air temperature is -10°C and wind speed is 10 m/s^{-1} .
- [Table 3.2](#) shows climatic data for McMurdo, Antarctica.
 - Calculate the mean annual temperature and the total annual rainfall.
 - Explain why higher temperatures occur between November and February.

▼ **Table 3.2** Climatic data for McMurdo, Antarctica

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
Temperature ($^{\circ}\text{C}$)	-2.9	-9.5	-18.2	-20.7	-21.7	-23.0	-25.7	-26.1	-24.1	-18.9	-9.7	-3.4	
Rainfall equivalent (mm)	15.0	21.2	24.1	18.4	23.7	24.9	15.6	11.3	11.8	9.7	9.5	15.7	

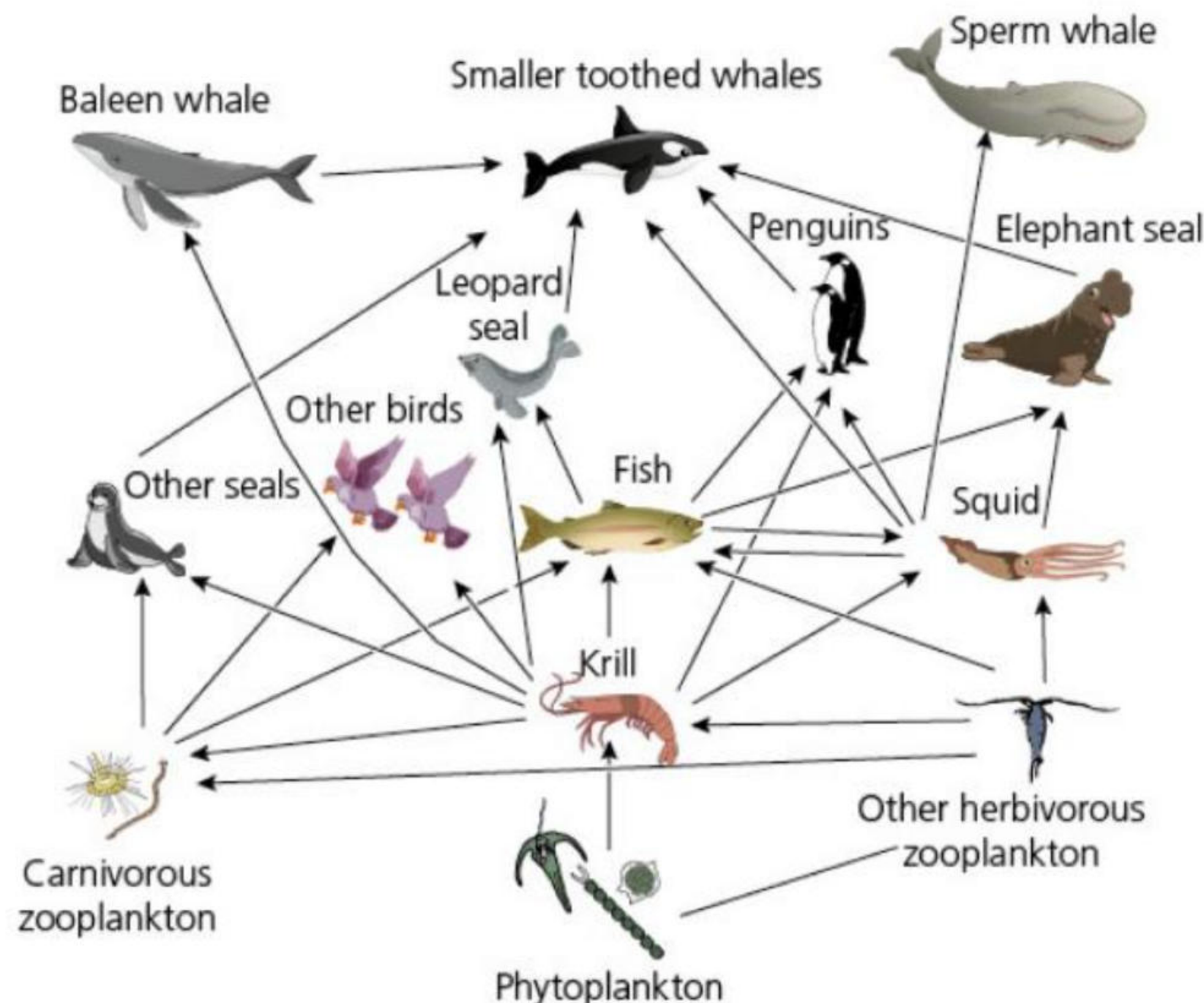
- Copy and complete the climate graph for McMurdo, Antarctica shown in [Figure 3.3](#).



▲ **Figure 3.3** Partially completed climate graph for McMurdo, Antarctica

The features of the Antarctic ecosystem

Despite the low temperatures, limited moisture, poor **soils** and lack of sunlight for six months, there are many **ecosystems** in and around Antarctica ([Figure 3.4](#)). There are around 200 species of lichen, more than 100 species of mosses and liverworts, 30 species of macrofungi, and large numbers of algae. Although the growing season is short, many species exist in the milder parts of the Antarctica Peninsula. Opportunities for photosynthesis and water availability are extremely limited, and growth rates and reproduction are very low, and some species, notably glass sponges in Terra Nova Bay on the Ross Sea, can live for up to 15,000 years.



▲ **Figure 3.4** Ecosystems around Antarctica

Antarctica's marine and aerial environments support an abundance of species, including penguins, seabirds, seals, whales and krill. The Southern Ocean contains some of the coldest sea water in the world. Cold water is very dense and sinks, displacing lower water to the surface. This upwelling water brings up oxygen and nutrients to the surface. This provides nutrients for surface algae, which then provide food for krill, a shrimp-like marine crustacean, which in turn provide food for fish, seals, whales and sea birds. Marine ecosystems in Antarctic waters may be relatively simple but they are, nonetheless, varied.

Krill ([Figure 3.5](#)) are about 5–6 cm long and feed on phytoplankton and zooplankton. Their biomass in Antarctic waters is estimated at around 500 million tonnes. Krill is commercially important, and between 150,000 and 200,000 krill are harvested in the Southern Ocean each year.



▲ **Figure 3.5** Krill

The Southern Ocean has over 250 species of fish that can survive temperatures below freezing, such as the Antarctic icefish. The Patagonian toothfish declined by about 50 per cent between 1985 and 2020 due to overfishing. Many species of whale are seasonal visitors to the Southern Ocean and these were hunted intensively in the early parts of the twentieth century. Species such as blue, humpback, minke, killer, southern right and sperm whales feed and breed around the Antarctic. Commercial harvesting of killer whales (orcas) last occurred in 1979–80. The population of orcas may now be in excess of 80,000. Likewise,

seals are found in large numbers in the Southern Ocean despite having previously been harvested. Seal species present include leopard, Ross, southern elephant and Weddell. Leopard seals can be over 3 m in length and weigh up to 500 kg. They feed on fish, penguins and other seals' pups. The Southern Ocean is home to nearly 90 per cent of benthic (deep sea) fauna.

Bird life includes terns, petrels, skuas, albatrosses and flightless penguins. Species of penguin include emperor, chinstrap, Adélie and royal. Emperor penguins are the largest – an adult male can be over 1 m tall and weigh around 40 kg. The emperor penguin is the only one to breed on the Antarctic Peninsula in winter. Albatrosses can have a wingspan of over 3.5 m. There are believed to be around 20,000 breeding pairs of albatrosses around the islands of the Southern Ocean.

Activities

- 1 Study [Figure 3.4](#). Create a food chain with seven levels in it.
 - 2 Suggest why food chains in the Antarctic oceans are more developed than food chains in the Antarctic continent.
-

The interrelationships between abiotic and biotic factors in the Antarctic ecosystem, and how flora and fauna adapt to survive

As elsewhere, abiotic factors in Antarctica include climate (for example temperature, precipitation and wind), nutrients, sunlight, soils, rocks and permafrost. Biotic factors include the mammals, birds, fish, insects, mosses and lichens found in the oceans, on land and in the soil.

There is a close relationship between abiotic factors and biotic factors. The lack of sunlight for six months, low temperatures, limited soil development and high wind speeds make life in central Antarctica difficult. However, around coastal areas, temperatures may be higher and conditions less extreme, allowing more species to survive. Many species use Antarctica for specific purposes, such as egg laying and raising their young.

In contrast, the oceans around Antarctica have a plentiful supply of nutrients at the surface due to the upwelling ocean currents. These support a large amount of plankton and krill, which in turn support large and diverse food webs that include fish, seals, penguins and whales.

Even sea ice can support ecosystems based on single-cell species such as bacteria and phytoplankton. These then support grazers such as zooplankton and krill, which in turn can support fish and other carnivores.

Life on land in Antarctica is somewhat limited. Most of the plant life includes mosses, lichen and liverworts, which support about 60 microspecies of invertebrates such as springtails and nematodes. The largest permanent dweller on Antarctica is the Antarctic midge (2–6 mm long).

There are many species adaptations to living in and around Antarctica. Seals, whales and penguins have thick layers of blubber (insulating fatty tissue). Blubber contains red blood cells that help regulate blood flow to the animals' skin surface. During warm conditions, the blood vessels expand and blood cells are taken to the surface. However, in cold conditions, blood vessels contract, taking blood cells away from the surface to vital organs.

Penguins have different types of feathers. On the surface there are coarse, abrasion-resistant, waterproof feathers. Over the course of a year these get damaged, so penguins replace these feathers each summer during a 'catastrophic moult'. Other feathers include soft, downy feathers, which provide insulation near the penguin's skin surface.

Penguins also form huddles. Emperor penguins may form large-scale huddles during major storms. The temperatures in the centre of the huddle can reach 37°C. Individuals cool down by moving towards the edge of the huddle and warm up by moving towards the centre.

Antarctic fur seals have two types of fur:

- » Coarse, long-guard hairs, which protect the under-fur from water and abrasion.
- » Dense (up to 50,000 hairs per cm²) water-resistant insulating fur.

Antarctic seals' eyes are adapted for hunting in dim conditions. They have large pupils that dilate to let more light in. Ross seals' eyes can be up to 7 cm in diameter. Seals also have very sensitive whiskers, which can detect the movement of prey at large distances.

Antarctic icefish have developed anti-freeze proteins in their blood. They are the only adult vertebrate with no red blood cells. The proteins bind to ice crystals, reducing their growth and preventing them from joining other ice crystals that could damage the icefish with their jagged edges.

Some species of sea spiders, worms and crustaceans reach very large sizes. It is believed that the high availability of oxygen in cold polar waters allows some species to keep on growing. This is known as gigantism.

Activities

- 1 Describe the relationship between abiotic factors and biotic factors in Antarctica.
 - 2 Explain how Antarctic fur seals are adapted to life in the Southern Ocean.
-

3.2 The threats to the Antarctic ecosystem and how they can be managed

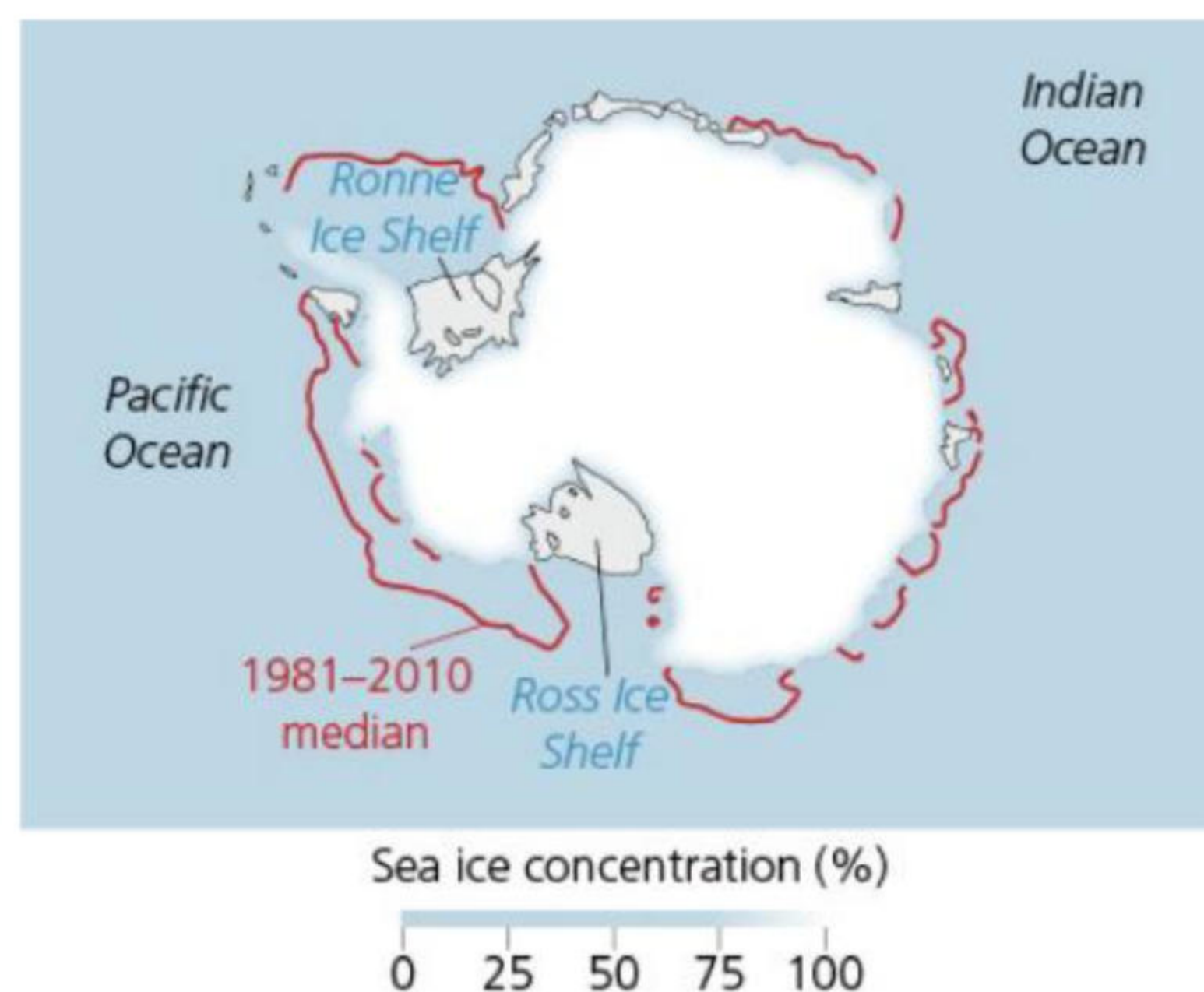
This chapter will explain:

- ★ the threats posed to Antarctica
- ★ the impacts of the destruction of the Antarctic ecosystem
- ★ to what extent the strategies and techniques used to manage the Antarctic ecosystem have been successful.

The threats posed to Antarctica

Climate change

Global climate change and the resultant melting ice sheets are major problems for Antarctica, in particular West Antarctica. In February 2023, Antarctic sea ice extent set a new record low ([Figure 3.6](#)), putting at risk the species that depend on stable sea ice as a breeding habitat. In 2023, Australian scientists reported that the strong ocean currents in the Southern Ocean could slow by more than 40 per cent by 2050. Moreover, the currents have already slowed by about 30 per cent since the 1990s. The reason is the melting taking place at the edge of Antarctica, which is adding more fresh water into the Southern Ocean, reducing its density. This reduces its ability to sink and to drive the deep flows that spread nutrients around the world.



▲ **Figure 3.6** The decline of ice around Antarctica, 1981–2010 compared with 2023

This has occurred with a global warming of about 1.1°C. If warming rises by 2°C – which is predicted by some to occur by 2100 – this would lead to major changes in the ocean and to the continent. It is not just ice at the edges that is melting, the whole ice cap is losing ice. Warmer water below the West Antarctica ice sheet is causing thinning from below.

Global warming is also having an impact on emperor penguins, as they breed and raise their young on sea ice. They are especially vulnerable to the loss of sea ice, to rising temperatures and altered wind systems. It is estimated that a rise in temperature of 1.5°C could lead to a decline of 19–31 per cent in emperor penguin colonies – but if global warming continues (business as usual) to 2100, up to 80 per cent of emperor penguin colonies could become extinct.

Resource exploitation

Mining and mineral exploitation are prohibited on Antarctica. However, scientific investigations into biotic resources are increasing. Some genetic and biochemical resources have been used to develop krill oil, food-related products and anti-freeze proteins. It is estimated that there are over 200 companies and research organisations involved in biological prospecting in Antarctica.

Fishing

Fishing in Antarctic waters is allowed and is increasing as fish stocks in other seas are increasingly overfished. Fishing practices can lead to a decline in predator species, increasing unintentional catch of juvenile fish, destruction of habitat by trawlers, and the death of non-target species caught up in nets and longlines. Longline fishing involves lengthy fishing lines with up to 20,000 baited hooks, which may also capture sea birds as they fish. Overfishing has led to the decline of species such as Antarctic hake and

Patagonian toothfish. Krill is endemic to Antarctic waters and many species depend on it as their main food source. However, krill is sensitive to rising ocean temperatures and it is also impacted by large-scale fishing and pollution in oceans.

Tourism

Antarctica receives upwards of 100,000 tourists per year. Transporting people to Antarctica involves emissions of 'black carbon' and other greenhouse gases by ships and aeroplanes. In addition, there is a risk of non-native insects, seeds and spores being carried in on vessels or by passengers. Most of the tourism pressure is on the Antarctic Peninsula, where 90 per cent of the tourism is concentrated.

Before the Covid-19 pandemic, some 74,000 tourists travelled to Antarctica, and around 20,000 remained on cruise ships visiting the continent. The numbers dropped to 23,000 during the pandemic but since then numbers have increased to around 100,000, with 75,000 'shore' visits.

Pollution

Pollution is an increasing problem. Microplastics are washed in from surrounding islands and seas, or they may be brought in by tourists or scientists. Sea birds often mistake plastic for food and eat it or feed it to their young. The remains of fishing equipment have been washed up on nearby islands. Seals and penguins are known to have become entangled in fishing nets, and the chances of rescue or release are slim.

Activities

- 1 Explain why the emperor penguin is vulnerable to global climate change.
 - 2 Explain why plastics and fishing equipment are a threat to Antarctica's ecosystems.
-

The impacts of the destruction of the Antarctic ecosystem

There has been continued human presence throughout the most biodiverse parts of Antarctica over the past 200 years. This has led to infrastructure developments, vegetation trampling, sealing, whaling, fishing, wildlife and soil disturbance, pollution and the introduction of alien species. However, human activity in Antarctica should be easier to regulate than elsewhere because it is restricted to science, television programmes and tourism, with few other human activities or permanent inhabitants.

Climate change is predicted to have major impacts on Antarctic ecosystems. For example:

- » Rapid warming in the west Antarctic Peninsula and a change from snowfall to summer rain has led to an increase in plant communities across the Antarctic Peninsula on newly exposed land, as well as the introduction of 'alien' organisms such as grasses, flies and bacteria.
- » Loss of sea ice west of the Antarctic Peninsula has led to a decline in krill and Adélie penguins. As krill is a major source of food for many marine animals and sea birds, this could lead to the decline of numbers of many species. The decline of sea ice means that species that rely on it are seeing their habitat shrink.
- » Models suggest a 3°C increase in temperature by 2100, which should increase phytoplankton productivity.

Historical changes are also important – for example, the whaling industry led to the near extinction of blue whales and baleen whales in the Southern Ocean. The extinction or near extinction of any single species has a knock-on effect on other species and contributes to loss of biodiversity. This can be seen here, as with less competition for krill from whales, the number of fur seals increased dramatically in just 20 years, from a few hundred to over 20,000. The seals came onshore at Signy Island in the South Orkney Islands, and caused much vegetation destruction through trampling, and eutrophication from the large amount of excrement left on the island.

Activities

- 1 Outline the impacts of global climate change on Antarctica's ecosystems.
 - 2 Explain how changes in the whaling industry led to ecosystem destruction on Signy Island.
-

An evaluation of the strategies and techniques used to manage the Antarctic ecosystem

International agreements

The main strategy for managing Antarctica has been the 1959 Antarctic Treaty. This established a system for international cooperation, especially in the name of science, and the demilitarisation of Antarctica. It also prohibits all forms of mining and mineral exploitation. This enables sustainable management of Antarctica's ecosystems.

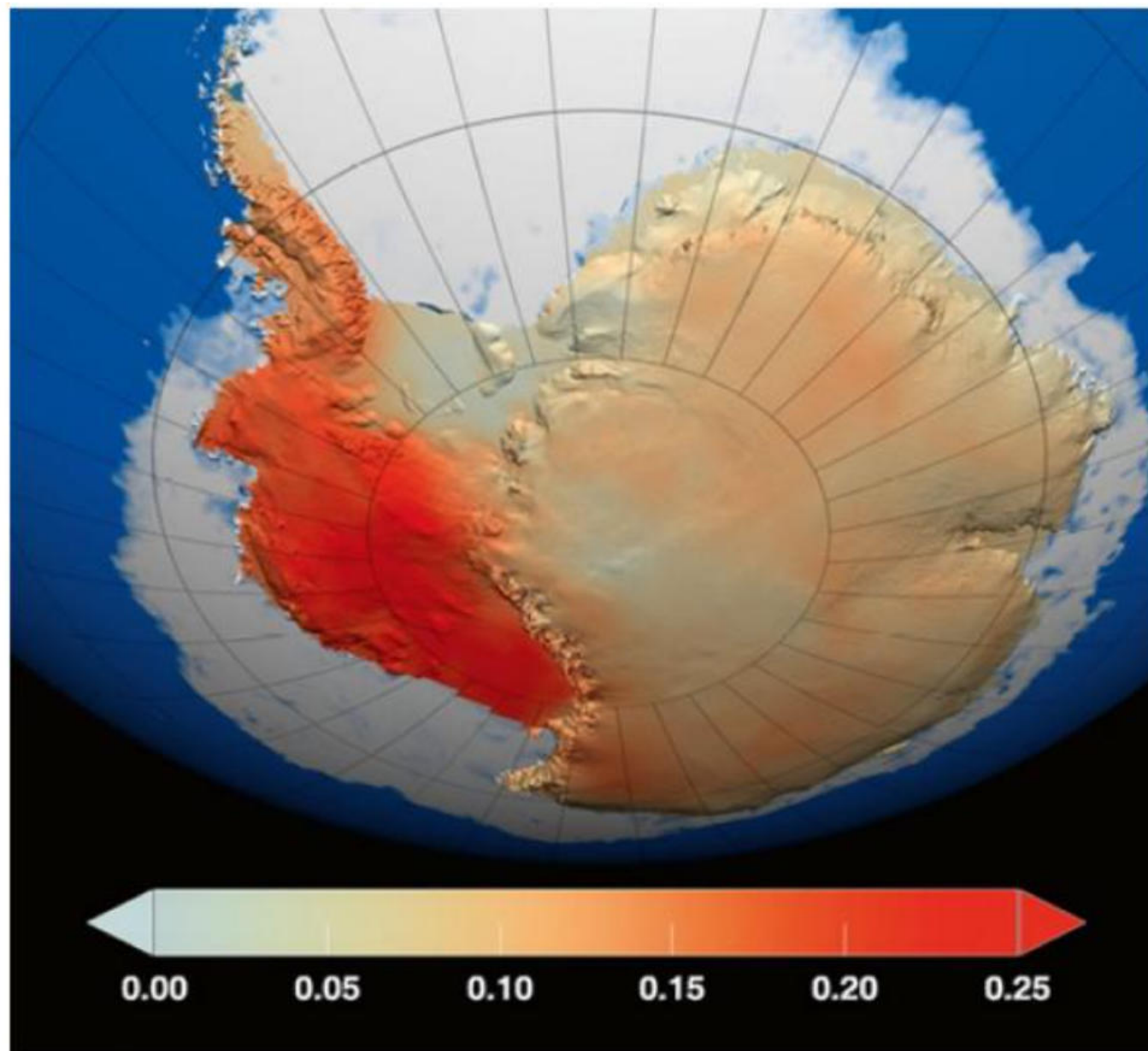
The principles and objectives of Article 9 of the 1959 Antarctic Treaty include:

- » the use of Antarctica for peaceful purposes only
- » the facilitation of scientific research in Antarctica

- » the facilitation of international scientific cooperation in Antarctica
- » the facilitation of the exercise of the right of inspection
- » questions relating to the exercise of jurisdiction in Antarctica
- » preservation and **conservation of living resources** in Antarctica.

The Treaty has been successful so far, partly due to the remoteness and isolation of Antarctica. Antarctica is the only continent on which there has not been a war. Its mineral resources have not been exploited, although its living natural resources such as whales, seals, fish and krill have been exploited. However, as Antarctica warms up and ice melts there could be increased pressure to develop its mineral resources.

Moreover, as global warming increases, the negative impacts on Antarctica are also likely to increase ([Figure 3.7](#)). Likewise, if the volume of tourists visiting the continent continues to increase, the impacts are likely to intensify. The same is true for pollution.



▲ **Figure 3.7** Antarctic warming since 1957 (°C)

Activities

- 1 Outline the main principles and objectives of the Antarctic Treaty.
- 2 Compare the potential impacts of low- and high-global warming emissions on Antarctica.

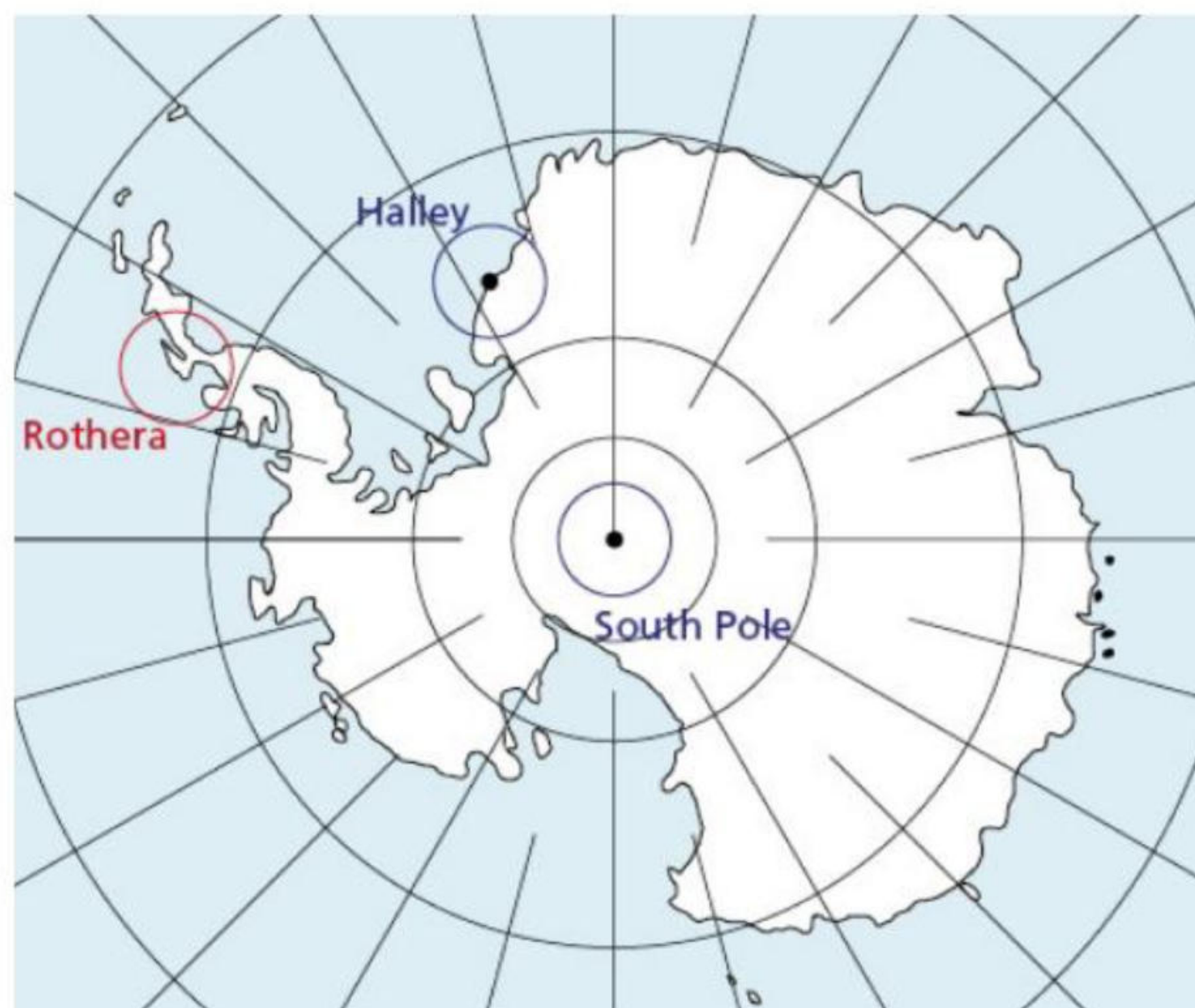
Environmental impact surveys

An **environmental impact survey** is a detailed survey to establish the likely impacts of a new development. Before any development starts, the environmental impact survey examines and identifies environments and species that are of high **conservation** importance or at risk from the development. These surveys make it possible to analyse the potential impacts of new developments on ecosystems and introduce management strategies to protect environments at risk.

By 2018, the wharf at the Rothera Research Station off the Antarctic Peninsula ([Figure 3.8](#)) was 25 years old and beyond repair. Moreover, more modern boats were longer and deeper than the wharf would allow to dock. The 'do nothing' and '**do minimum**' approaches were considered unsafe for the berthing of vessels, so a new wharf had to be constructed.



▲ **Figure 3.8** Rothera Research Station



▲ **Figure 3.9** The location of the Rothera Research Station, Antarctica

Some 24,000–30,000 m³ of local rock was sourced for the new wharf. Local material had to be used as the risk of introducing alien species was too high. Levels of biodiversity at Rothera were not high. Although seals were present, they did not breed in the area. The most significant environmental impacts were:

- » the introduction of non-native species
- » terrestrial and marine pollution from oil spills
- » the removal of rock
- » the removal of ice-free areas for terrestrial habitats
- » the disturbance of marine mammals by underwater noise.

The wharf was started in 2018 and completed in 2020.

Activities

- 1 Define the term 'environmental impact survey'.

2 Outline the potential impacts of new buildings on Antarctica’s environment.

3.3 The characteristics of the tropical rainforest ecosystem

This chapter will explain:

- ★ the global distribution of tropical rainforests
- ★ the characteristics of the equatorial climate
- ★ the characteristics and structure of the tropical rainforest ecosystem.

The global distribution of tropical rainforests

Evergreen **tropical rainforests** are located in equatorial areas, largely between 10°N and 10°S ([Figure 3.10](#)). There are some areas of rainforest found outside these regions, but these tend to be more seasonal in nature. The main areas of rainforest include the Amazon rainforest in Brazil, the Congo rainforest in central Africa and the Indonesian–Malaysian rainforests of Southeast Asia. There are also many small fragments of rainforest, such as those on the island of Madagascar and in the Caribbean. Tropical rainforests everywhere are under increasing threat from human activities such as farming and logging. The result is that rainforests are disappearing and those that remain are not only smaller, but are broken up into fragments.

Interesting note

Tropical rainforests cover 6 per cent of the world's land surface but hold 50 per cent of the world's species. The Amazon rainforest alone is home to 10 per cent of the world's known species.



▲ **Figure 3.10** World distribution of tropical rainforests

The characteristics of the equatorial climate

The main characteristics of an equatorial climate include:

- » hot conditions – generally above 26°C – throughout the year, as the overhead sun is always high in the sky by the middle of the day
- » high levels of rainfall, often over 2000 mm ([Figure 3.11](#))
- » tropical rainforests characterised by low pressure – rising air that cools and produces clouds and rains
- » a lack of seasons – the temperatures are high throughout the year
- » the difference between daytime and night-time temperatures (known as the diurnal range) is higher than the seasonal differences in temperature
- » rainfall is mainly convectional – heating causes air to rise and then cool, and rain may fall on as many as 250 days each year
- » cloud cover varies – in the morning it may be limited, but by the afternoon towering cumulonimbus clouds mark the start of the **convectional rains**
- » the presence of clouds tends to reduce the amount of heat that is lost at night – hence the diurnal range is less than in hot desert areas
- » the **humidity** (moisture in the atmosphere) is high, and relative humidities of 100 per cent are often

reached in the late afternoon

- » wind speeds within the rainforest are reduced by the large numbers of trees present
- » with increasing latitude there is increased seasonality of the rainforest.



▲ **Figure 3.11** Convictional uplift causes cloud and rain on most days in a tropical rainforest, such as here in Sarawak

The data for Manaus in Brazil ([Table 3.3](#)) show that the warmest months are September and October, with a maximum temperature of 34°C. In contrast, all of the months from December to September share the minimum temperature of 24°C. Thus the annual temperature range is 10°C.

Rainfall in Manaus is high, nearly 2100 mm annually. There is a definite wet season between November and May, whereas the months of June to October are relatively dry.

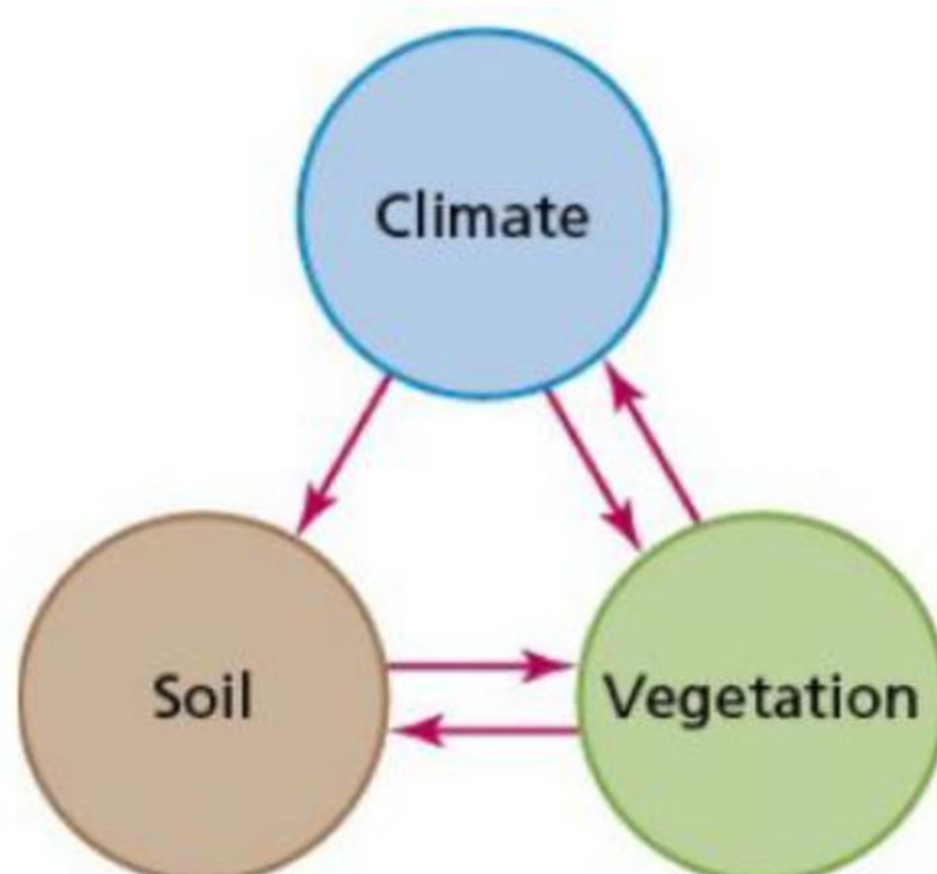
Activities

- 1 Calculate for Manaus:
 - a the mean average monthly temperature
 - b the annual precipitation.
 - 2 Compare the seasonal variations in rainfall and sunshine (hours) for Manaus.
-

The characteristics and structure of the tropical rainforest ecosystem

The tropical rainforest ecosystem includes biotic factors (such as plants and animals) and abiotic factors (such as heat, rainfall, oxygen and minerals). These are closely interrelated ([Figure 3.12](#)).

The links between climate, soil and vegetation are very strong



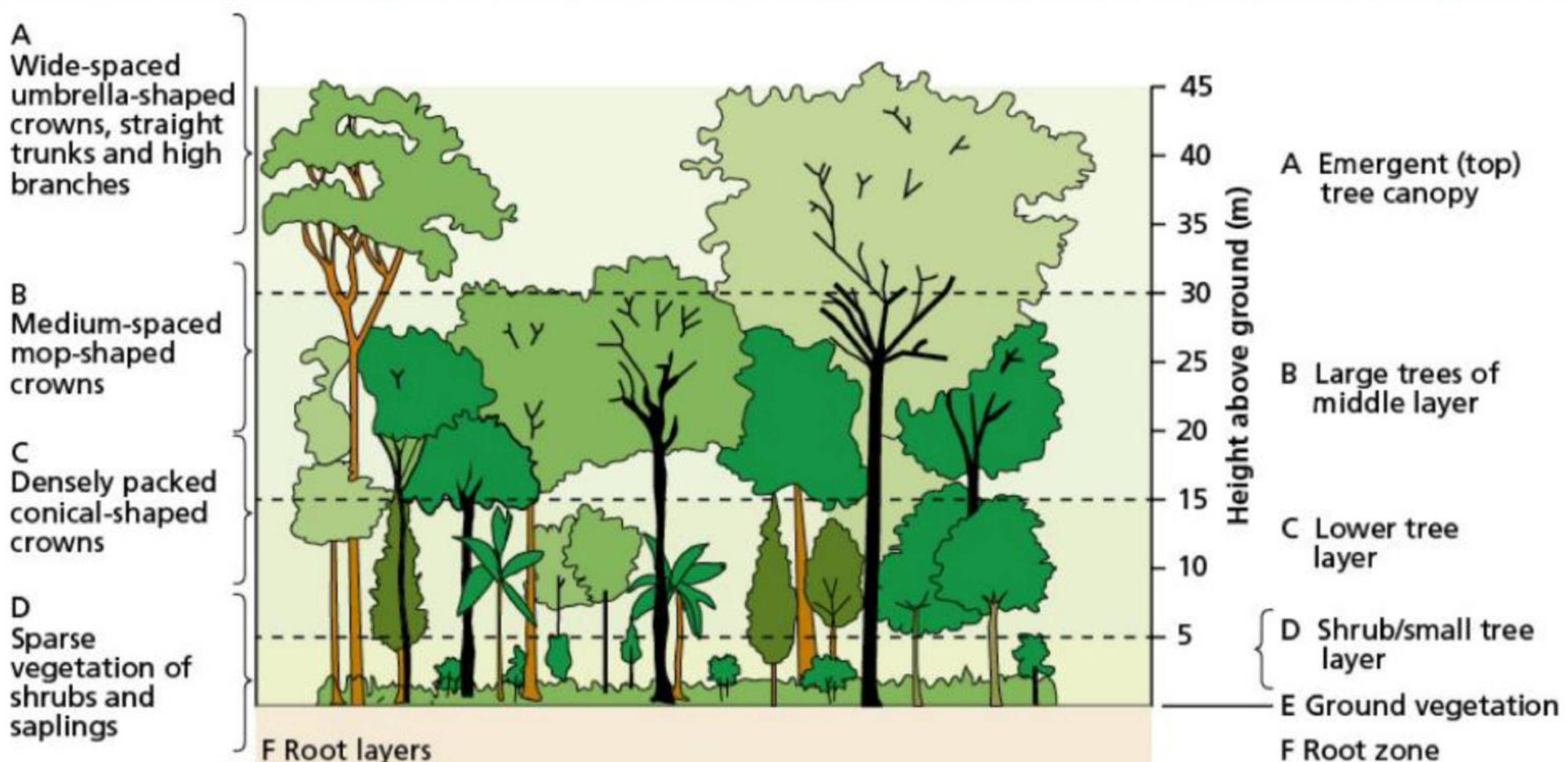
▲ **Figure 3.12** The links between climate, soil and vegetation in a tropical rainforest ecosystem

Vegetation is evergreen, enabling photosynthesis to take place all year round. This is possible due to the high temperatures and presence of water throughout the year. The vegetation is layered, and the shape of the crowns varies at each layer ([Figure 3.13](#)). Species at the top of the canopy receive most of the sunlight, whereas species that are located near the forest floor are adapted to darker conditions and generally have a darker pigment so as to photosynthesise at low light levels. There is a great variety in the number of species in a rainforest – this is known as **biodiversity**. A single hectare in a rainforest may contain as

many as 300 different species. Typical rainforest species include figs, teak, mahogany and yellow woods.

▼ **Table 3.3** Climate data for Manaus, Brazil

	J	F	M	A	M	J	J	A	S	O	N	D	Av/Total
Temperature:													
Daily max (°C)	31	31	31	31	31	31	32	33	34	34	33	32	32
Daily min (°C)	24	24	24	24	24	24	24	24	24	25	25	24	24
Average monthly (°C)	28	28	28	27	28	28	28	29	29	29	29	28	
Rainfall:													
Monthly total (mm)	278	278	300	287	193	99	61	41	62	112	165	220	
Sunshine:													
Sunshine (hours)	3.9	4	3.6	3.9	5.4	6.9	7.9	8.2	7.5	6.6	5.9	4.9	5.7



▲ **Figure 3.13** Vegetation structure of the tropical rainforest

Soils

Rainforests are the most productive land-based ecosystems. Ironically, the soils of tropical rainforests are quite infertile. This is because most of the nutrients in the rainforest are contained in the biomass (living matter). Rainforest soils are typically deep due to the large amount of weathering that has taken place, and they are often red in colour due to large amounts of iron being present in the soil. Nevertheless, there are some areas in which tropical soils may be more fertile: in floodplains and in volcanic areas the soils may be enriched by flooding or by the weathering of fertile lava flows.

How flora and fauna adapt to survive

Flora

Tropical vegetation in rainforests has many adaptations. Some trees have leaves with drip-tips ([Figure 3.14a](#)), which are designed to get rid of excess moisture. In contrast, other plants have saucer-shaped leaves in order to collect water. Pitcher plants have developed an unusual means of getting their nutrients. Rather than taking nutrients from the soil, they have become carnivorous and get their nutrients from insects and small frogs that are trapped inside the pitcher ([Figure 3.14b](#)). This is one way of coping with the very infertile soils of the rainforest. Other plants are very tall. To prevent being blown over by the wind, very large trees have developed buttress roots that project out from the main trunk above the ground, which gives the plant extra leverage in the wind.



▲ **Figure 3.14** Adaptations of rainforest plants: a drip-tip; b pitcher plant

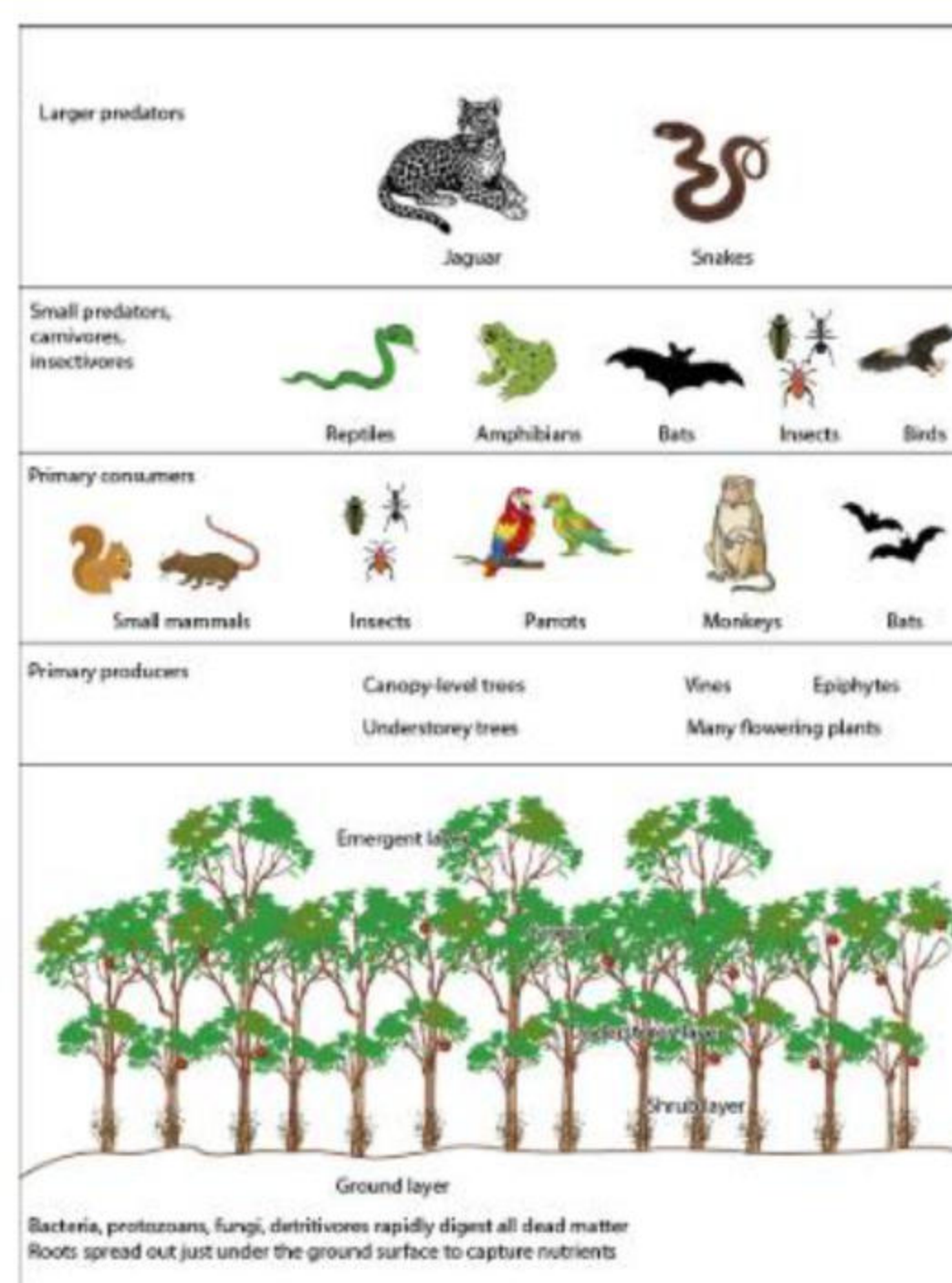
Fauna

Although rainforests cover less than 6 per cent of the Earth's surface, they account for over 50 per cent of all animal species on Earth. Tropical rainforests are very biodiverse ([Figure 3.15](#)). Many species, such as orangutans, are arboreal (they live entirely in the trees) and rarely come to the ground, as they would be easy prey for large carnivores there.

Top carnivores, such as jaguars, tigers and leopards, are highly camouflaged to aid in hunting. Many other species are highly camouflaged to avoid becoming prey, such as stick insects and the Indian oakleaf butterfly. Sloths are covered with a layer of green algae that camouflages their fur in their arboreal environment. Some animals have evolved to look larger or more scary than they really are: the larvae of the lobster moth look like scorpions, but they are defenceless. Many butterflies have designs that look like large eyes on their wings, in order to confuse potential predators. Some species, such as the monarch butterfly, are poisonous, which helps deter predators. It is more than likely that there are many rainforest species that have not yet been discovered – according to the World Wide Fund for Nature, new species are being discovered in the Amazon 'every other day'. Between 2016 and 2020, over 600 new species of plants and animals were discovered in the Amazon alone.

Activities

- 1 Suggest how the vegetation in [Figure 3.15](#) is adapted to conditions in the rainforest.
- 2 Explain how animals are adapted to living in tropical rainforests.



▲ **Figure 3.15** A simplified food chain from the tropical rainforest

3.4 The threats to the tropical rainforest ecosystem and how they can be managed

This chapter will explain:

- ★ the main threats to tropical rainforest ecosystems
- ★ the impacts and causes of deforestation in the Brazilian Amazonian rainforest
- ★ to what extent the strategies and techniques used to manage tropical rainforest ecosystems have been successful
- ★ detailed specific example: Danum Valley Conservation Area, Borneo.

The main threats to tropical rainforest ecosystems

About 200 million people live in areas that are or were covered by tropical rainforests. These areas offer many advantages for human activities, such as farming, hydroelectric power, tourism, fishing and food supply, mineral development and forestry ([Figure 3.16](#) and [Table 3.4](#)). Rainforests also play a vital role in regulating the world’s climate, and they account for 50 per cent of the world’s plants and animals. They are vital too for the protection of soil and water resources.

▼ **Table 3.4** The value of tropical rainforests

Industrial uses	Ecological uses	Subsistence uses
● Charcoal ● Saw logs ● Gums, resins and oils ● Pulpwood ● Plywood and veneer ● Industrial chemicals ● Medicines ● Genes for crops ● Tourism	● Watershed protection ● Flood and landslide protection ● Soil erosion control ● Climate regulation, e.g. balancing levels of carbon dioxide and oxygen ● Special woods and ashes ● Fruit and nuts	● Fuelwood and charcoal ● Fodder for agriculture ● Building poles ● Pit sawing and saw milling ● Weaving materials and dyes ● Rearing silkworms and beekeeping



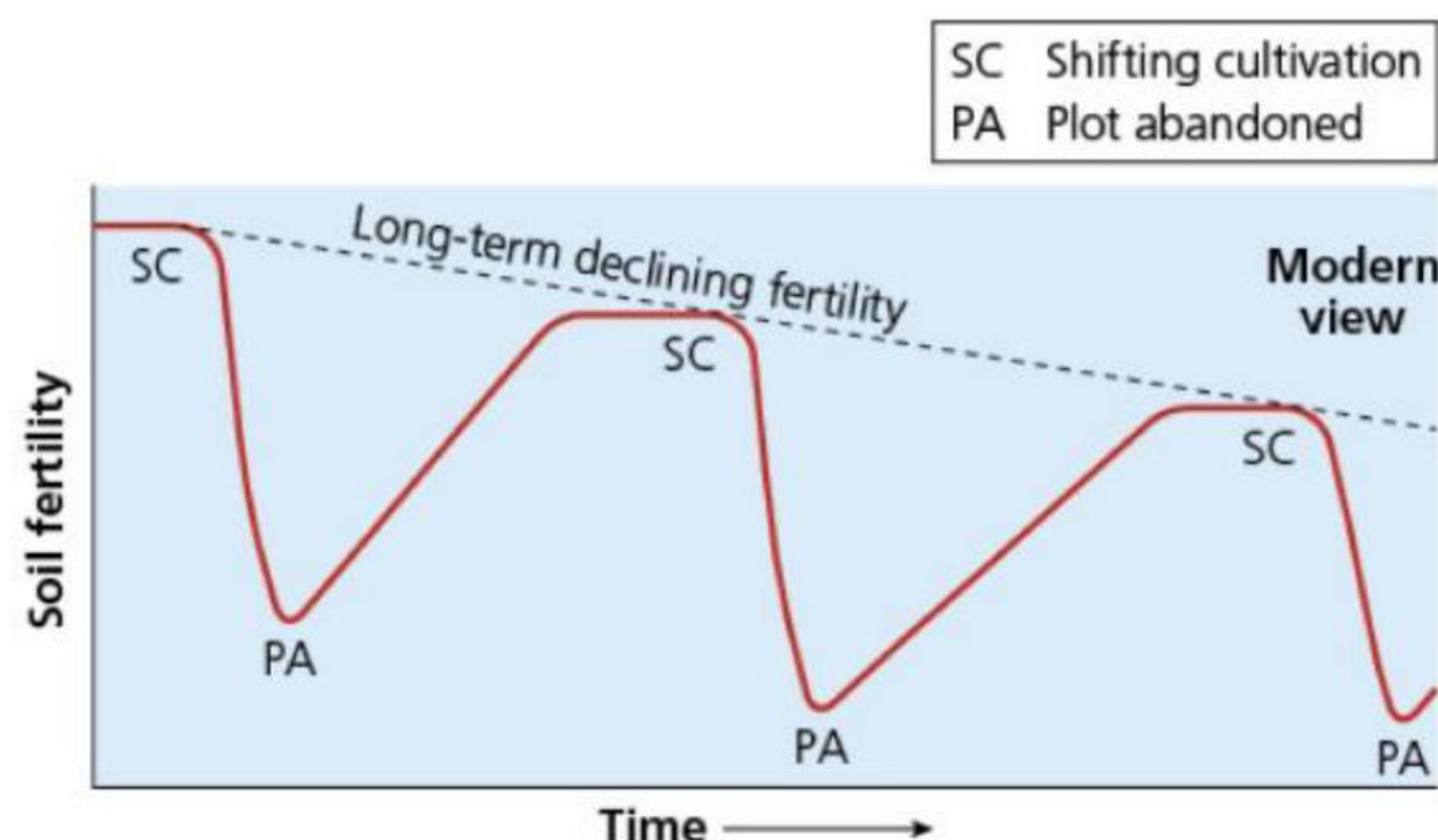
▲ **Figure 3.16** A tropical rainforest along with shifting cultivation – rice growing in Sarawak

However, tropical rainforests are under threat:

- » The year-round growing season is very attractive for farmers, although the poor quality of the soil results in the land being farmed for only a few years before it is abandoned ([Figure 3.17](#)). Nevertheless, large-scale **plantations** occur in areas of tropical rainforest, producing crops such as palm oil, which is increasingly being used for the **biofuels** industry.
- » Cattle ranching is a threat to rainforests as it replaces a multi-layered forest ecosystem, which intercepts most of the rainfall, with a single-layered grassland ecosystem, which gets trampled by cattle and can

produce much overland runoff and soil erosion.

- » The construction of roads and railways has two main impacts. The first is direct – rainforests are cut down to make way for roads, such as the Trans-Amazonian Highway. As a result of the increased accessibility provided by the new roads and railways, some settlers will cut down more rainforest to produce farmland.



▲ **Figure 3.17** The effect of rainforest clearance on long-term soil fertility

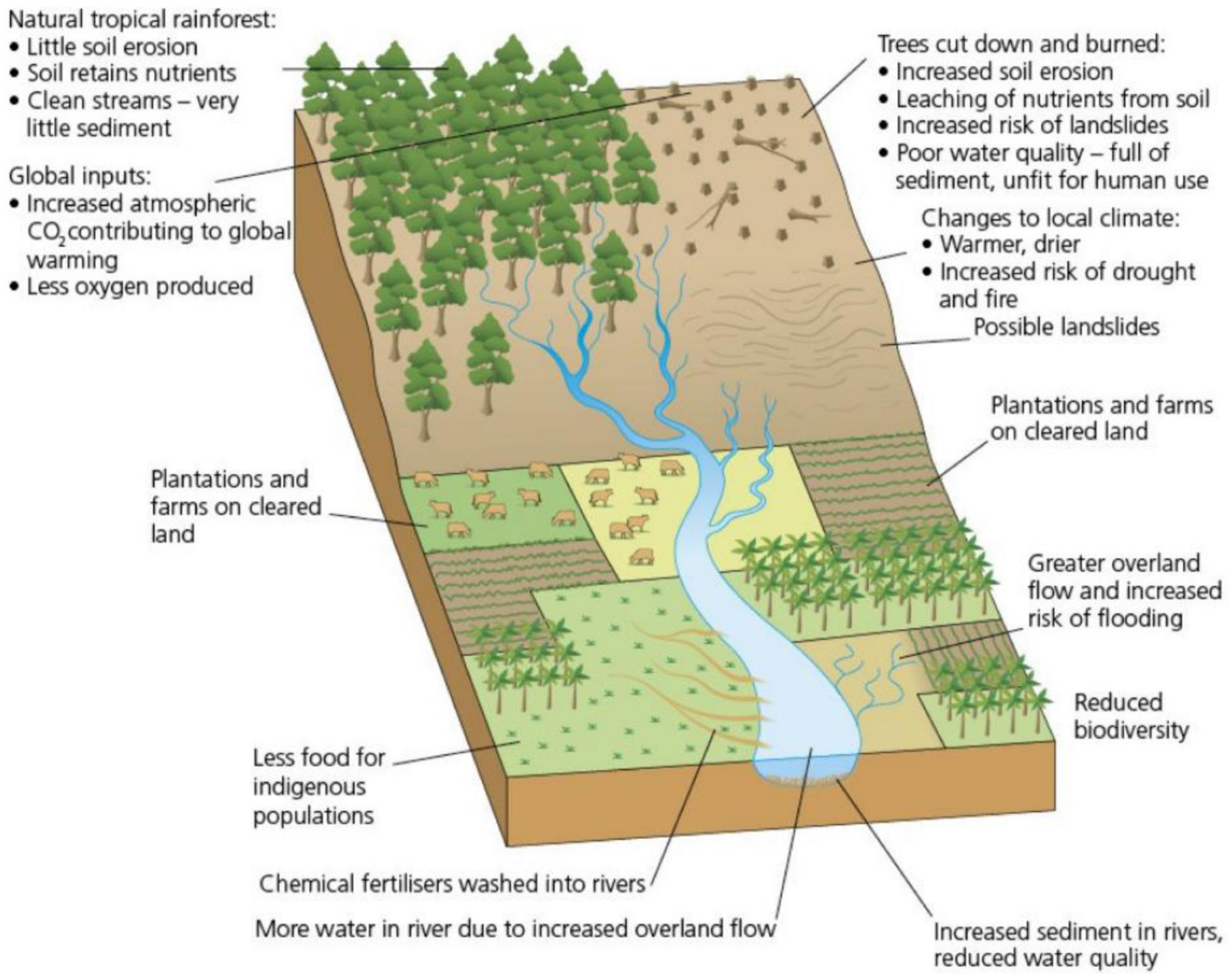
- » The second impact of the construction of roads and railways is that settlements will begin to develop along these roads and railways to provide more services for the new settlers. Each of these developments leads to a decline in the area of tropical rainforest and the services that it provides to people.
- » High rainfall totals, especially in hilly areas, favour the development of hydroelectric power (HEP), such as at Batang Ai in Sarawak, Malaysia ([Figure 3.19](#)).
- » Areas of rainforest have a long history of commercial logging. Tropical hardwoods, such as teak and mahogany, are prized by furniture manufacturers.
- » Mineral developments, such as iron ore at Carajas in Brazil and ilmenite on the southeast coast of Madagascar, are also developed in some rainforest areas and these resources may be exploited.

The impacts and causes of deforestation in the Brazilian Amazonian rainforest

Impacts

There are a large number of effects of deforestation ([Figure 3.18](#)), including:

- » disruption to the circulation and storage of nutrients
- » surface erosion and compaction of soils
- » increased flood levels and sediment content of rivers
- » climatic change
- » loss of biodiversity and animal habitats.



▲ **Figure 3.18** Some impacts of deforestation



▲ **Figure 3.19** The rainforest at Batang Ai, affected by flooding, shifting cultivation and soil erosion

Deforestation disrupts the closed system of **nutrient cycling** within tropical rainforests. Inorganic elements are released through burning and are quickly flushed out of the system by the high intensity rains.

Soil erosion is also associated with deforestation. As a result of soil compaction, there is a decrease in infiltration, an increase in overland runoff and surface erosion.

As a result of the intense surface runoff and soil erosion, rivers have a higher **flood peak** and a shorter time lag. However, in the dry season river levels are lower, the rivers have greater turbidity (murkiness due to more sediment), an increased bed load, and carry more silt and clay in suspension.

Other changes relate to **climate**. As deforestation progresses, there is a reduction of water that is re-evaporated from the vegetation, hence the recycling of water is diminished. Evaporation rates from savanna grasslands are estimated to be only about one-third of that of the tropical rainforest. Thus, mean annual rainfall is reduced, and the seasonality of rainfall increases.

Causes

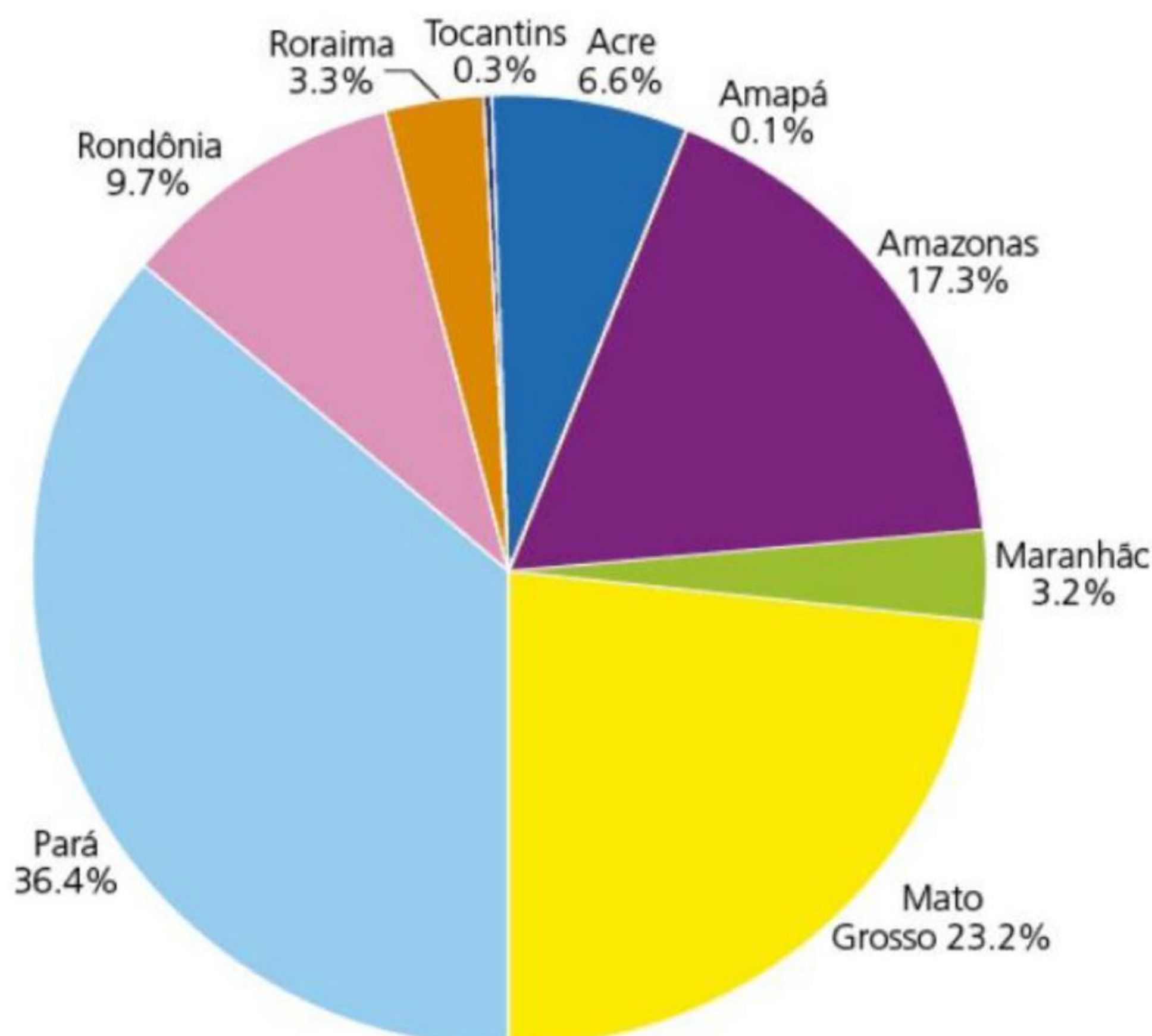
There are six main causes of deforestation in Brazil:

- » Agricultural colonisation by landless migrants and speculative developers along highways and agricultural growth areas.
- » Conversion of the forest to cattle pastures, especially in eastern and southeastern Para and northern Mato Grosso.
- » Mining, for example the Greater Carajas Project in southeastern Amazonia, which includes a 900 km railway and extensive deforestation to provide charcoal to smelt the iron ore; another threat from mining comes from the small-scale informal gold mines, *garimpeiros*, causing localised deforestation and contaminated water supplies.
- » Large-scale hydroelectric power schemes such as the Tucuruí Dam on the Tocantins River.
- » Forestry taking place in Para, Amazonas and northern Mato Grosso.
- » Use of the forest by local populations is a necessity in order to earn a living and have an acceptable quality of life.

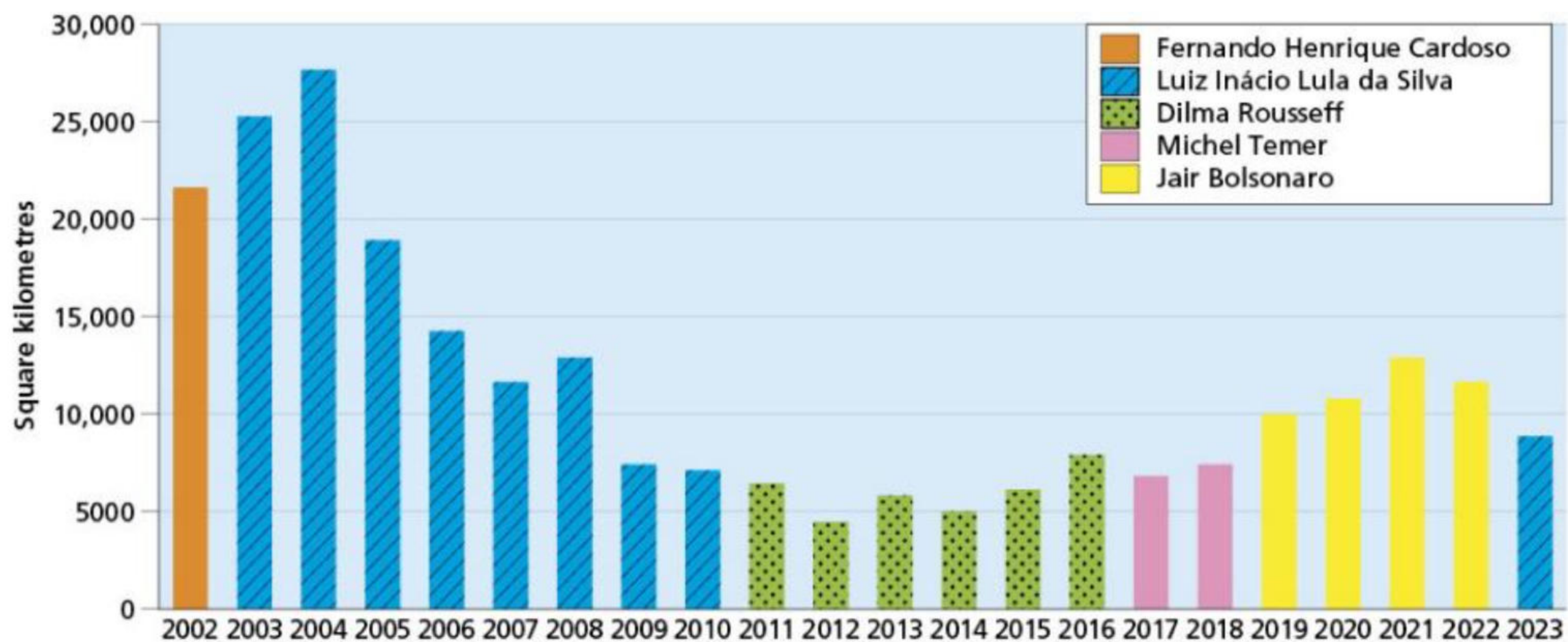
Deforestation in Brazil shows several trends:

- » It is a recent phenomenon.
- » It has partly been promoted by government policies.
- » There is a wide range of causes of the deforestation.
- » It includes new areas of deforestation as well as the extension of previously deforested areas.
- » A major cause of the deforestation is land speculation and the granting of land titles to those who 'occupy' parts of the rainforest.

According to the news organisation Mongabay, in 2023 deforestation from conversion of forest to soy farming continued to spread through Brazil's Cerrado and Amazon rainforest. There was around 27,000 ha of deforestation and forest degradation in the Cerrado in just four months, while the Amazon saw over 30,000 ha. All of it was located near grain silos used by the seven biggest soy traders in Brazil ([Figures 3.20](#) and [3.21](#)).



▲ **Figure 3.20** Deforestation of the Amazon by state, 2023



▲ **Figure 3.21** Deforestation of the Amazon under the president in office at the time, 2002–23

[Figure 3.20](#) shows that most of the deforestation occurred in just two states, Pará and Mato Grosso. However, the long-term trend in deforestation is going down. The current decline is largely due to Brazilian President Lula, who aims to eliminate illegal deforestation by 2030. This contrasts with the significant increase in deforestation under the previous president, Jair Bolsonaro.

Activities

- 1 Comment on the value of tropical rainforests for human populations.
- 2 Outline the main impacts of deforestation on the natural environment.
- 3 Explain the main causes of deforestation in Brazil.
- 4 Comment on the trends in deforestation in Brazil.

The strategies and techniques used to manage tropical rainforest ecosystems

Because tropical rainforests are important for many services, such as flood control, climate regulation, a source of genetic material, timber and food, and many others ([Table 3.4, page 80](#)), attempts have been made to protect rainforests. An example of the sustainable management of tropical rainforests is the Heart of Borneo (HoB) project. The governments of Malaysia, Indonesia and Brunei created the HoB, an area of over 20 million ha, to sustainably protect the rainforest and water resources for the welfare of present and future generations. It protects forest cover, manages protected areas and maintains biodiversity and environmental services in productive activities such as farming, forestry, fishing and ecotourism. Nevertheless, many areas of rainforest are still being cut down, in part due to population growth and the need to find more food supplies, but also due to people who want to cut down rainforests to get at mineral resources below ground.

Elsewhere in Borneo, over two-thirds of Borneo's dipterocarps (tall, straight trees providing high quality timber) are threatened with extinction. Up to 350 species of mammals, birds, reptiles and fish are at risk of extinction, for example clouded leopards, orangutans and Sumatran rhinoceroses. Other impacts include habitat losses, carbon emissions, fires (used to clear forests), land degradation and decreased water quality – palm oil plantations often use chemical fertilisers and pesticides which can pollute rivers and groundwater. Land drainage increases soil erosion. There are also social issues as the palm oil industry often removes indigenous people from their land.

Detailed specific example

Danum Valley Conservation Area, Malaysian Borneo

The Danum Valley Conservation Area (DVCA) in Borneo contains more than 120 mammal species, including ten species of primate. The DVCA and surrounding forest is an important reservation for orangutans. These forests are particularly rich in other large mammals, including the Asian elephant, Malayan sun bear, clouded leopard, bearded pig and several species of deer. The area also provides one of the last refuges in Sabah, northeast Borneo, for the critically endangered Sumatran rhino. Over 340 species of bird have been recorded at Danum, including the argus pheasant, Bulwer's pheasant, and seven species of pitta bird.

The DVCA covers 43,800 ha, comprising almost entirely lowland dipterocarp forest (dipterocarps are valuable hardwood trees). It is the largest expanse of pristine forest of this type remaining in Sabah ([Figure 3.22](#)).

Until the late 1980s, the area was under threat from commercial logging. The establishment of a long-term research programme between Yayasan Sabah and the Royal Society in the UK created local awareness of the conservation value of the area and provided important scientific information about the forest and what happens to it when it is disturbed through logging. Danum Valley is controlled by a management committee containing all the relevant local institutions –

the wildlife, forestry and commercial sectors are all represented. This allows Danum Valley to be managed sustainably. To the east of the DVCA is the 30,000 ha Innoprise-FACE Foundation Rainforest Rehabilitation Project (INFAPRO), one of the largest forest rehabilitation projects in Southeast Asia, which is replanting areas of heavily disturbed logged forest.

Because all areas of conservation and replanting are embedded within the larger commercial forest, the value of the whole area is greatly enhanced. Movement of animals between forest areas is enabled and allows the continued survival of some important and endangered animals such as the Sumatran rhino, the orangutan and the Borneo elephant.

In the late 1990s, a hotel was established on the northeastern edge of the DVCA. It has established flourishing ecotourism in the area and exposed this unique forest to a wider range of visitors than was previously possible. As well as raising revenue for the local area, it has raised the international profile of the region as an important centre for conservation and research. This is a very good example of the sustainable development of an area of tropical rainforest.

Activities

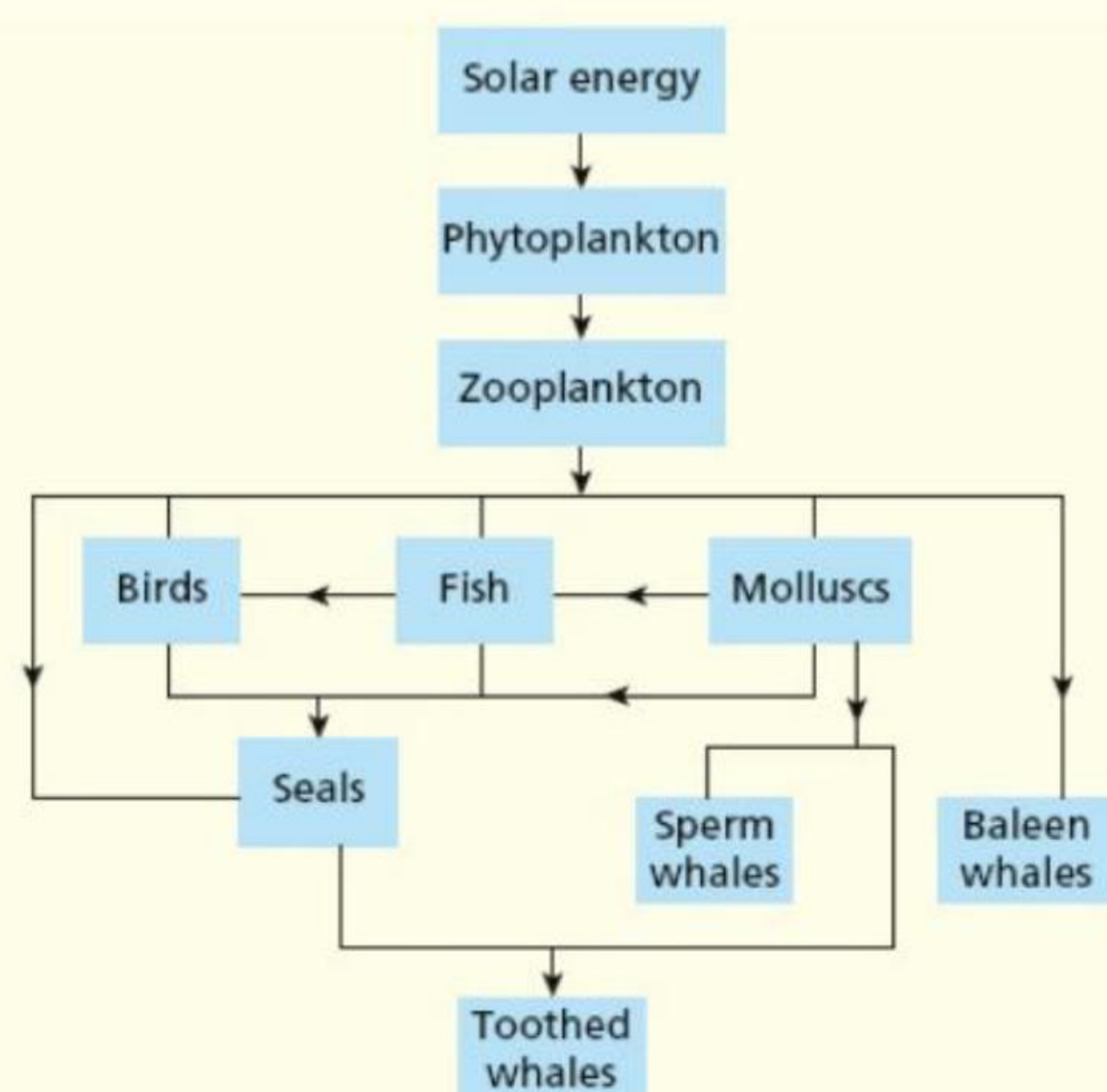
- 1 What was the main threat to the Danum Valley before the late 1980s?
- 2 Why is the DVCA important for the conservation of species?



▲ **Figure 3.22** The Danum Valley Conservation Area

Practice questions

- 1 [Figure 3.23](#) shows a food web for the sea around Antarctica.



▲ **Figure 3.23** A food web for the sea around Antarctica

- a**
 - i** Identify the main source of energy for this ecosystem. [1]
 - ii** Identify the organism that feeds on phytoplankton. [1]
 - iii** State the organism that feeds on birds. [1]
 - iv** State the top carnivore in the Antarctic marine ecosystem. [1]
 - b** Explain why Antarctica's ecosystems are largely marine-based rather than land-based. [3]
- 2** View [Figure 3.16 on page 80](#) which shows an area of tropical rainforest.
- a** Identify how this area of rainforest is being used. [1]
 - b** Compare the characteristics of the vegetation in the rainforest with that of the cleared area. [2]
 - c** Explain two reasons why tropical rainforests are being destroyed by human activities. [4]
 - d** Explain the consequences of the deforestation of tropical rainforests. [4]